

Proactive Intervention Desk

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1 Introduction

A Multi-Agent System (MAS) with all agents being ambient needs to be proposed. I learned about MAS and its architecture through the following resources:

- Multi-Agent Systems (MAS)
- Ambient Agent
- How Salesforce manages its solutions
- Memory in AI
- Memory in Multi-Agent Systems
- Connect Everything with Agentforce MuleSoft
- What is Salesforce
- Model-Agnostic Hooks
- Multi-Agent Systems: A Paradigm to Design Ambient Intelligent Applications
- Query on Non-Negotiables

2 Agent Memory

Working Memory: Per-customer, session-scoped, and discarded once the session is completed.

Episodic Memory: Per-customer memory that persists interventions, outcomes, and the current life-phase field. This is the one thing that the specification explicitly says must survive across weeks and be read by multiple agents such as Offer, Retention, and Synthesis, rather than being re-derived.

Semantic Memory: Shared across customers; eligibility rules and current compliance policy. A2A can be used to communicate between agents.

3 MAS Design

Swarm: Applied to Usage, Transaction, and Support Agents. They can function simultaneously because they watch the same customer at once, independently, without waiting on each other. They run in parallel, and each publishes its structured conclusion to the shared per-customer state board.

Agent Debate: Specifically inside Synthesis, when swarm findings conflict. The debate output gets logged as a resolved-or-escalated state, not silently averaged.

Agent-Dependent Handoff: Used for Life-Event Inference and Synthesis. These should not fire on every raw event, only once a correlated signal exists.

Critique-Refiner: The mandatory last LLM step before anything reaches a human reviewer. It reviews cost feasibility and compliance.

Round-Robin: Used for drafting any customer-facing message.

Guardrail: Deterministic keyword/classifier match and hard-coded bypass. It can route straight to a legal/fraud queue without waiting for any agent’s reasoning.

4 Agent and Trigger Types

Agent	Trigger Type	Trigger
Usage	Event + time-based	Each usage event; nightly rollup
Support	Event-based	New ticket or message
Transaction	Event-based	Real-time transaction stream
KYC	Event + time-based	KYC update; periodic re-verification
Guardrail	Event-based, independent	Legal/fraud/self-harm keyword match – bypasses everything
Life-Event Inference	Agent-dependent	Correlated signals from upstream agents
Synthesis	Agent-dependent	Once swarm agents have published to the state board
Offer/Eligibility	Agent-dependent	Synthesis produces a candidate opportunity
Retention/Action	Agent-dependent	Synthesis + Offer agree there is a case; this agent selects one of the six bounded actions
Critique-Refiner	Agent-dependent	Immediately after the Action agent drafts a proposal

Table 1: Agents and their corresponding trigger types

5 Flowchart

The proposed architecture of the ambient multi-agent system is shown in Figure 1.

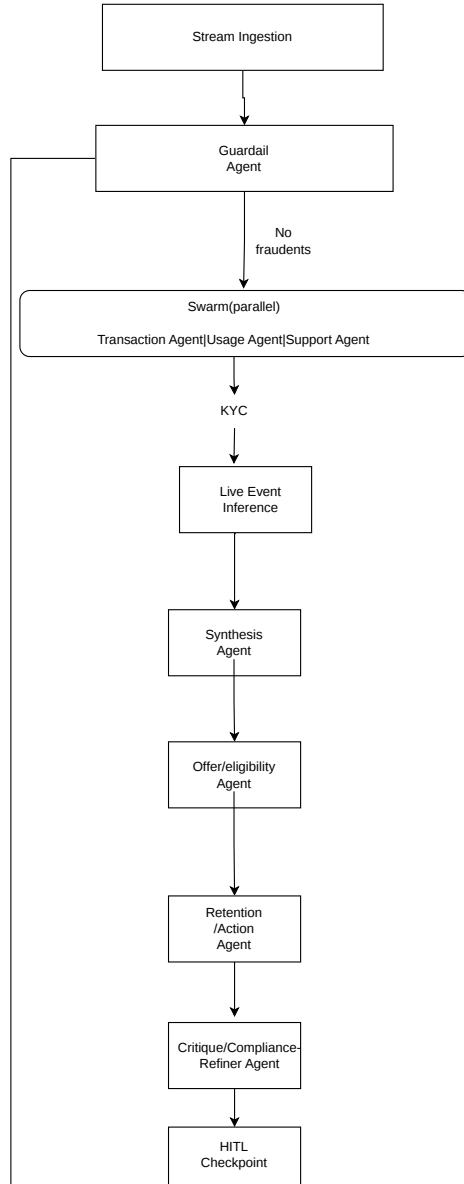


Figure 1: Proposed Ambient Multi-Agent System Architecture